

Exploratory Data Analysis Report

Seismic First-Break Picking Dataset Evaluation

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1. Executive Summary & Objective

This report provides a comprehensive exploratory data analysis of the seismic survey file to establish a ground-truth understanding prior to integration. [cite: 1] Because the primary objective involves detecting first break arrival times utilizing U-Net segmentation models, the survey must be properly verified to construct accurate training masks and configure HDF5 dataset loaders appropriately. [cite: 1] The dataset aims to facilitate the segmentation of each shot gather into before-break, break, and after-break regions. [cite: 1]

Target File: Lalor_raw_z_1500ms_norp_geom_v3.hdf5 [cite: 1]

File Size: 15.67 GB on disk. [cite: 1]

Structure: Contains 108 internal HDF5 objects (groups and datasets), organized primarily under the TRACE_DATA/DEFAULT group. [cite: 1]

2. Structural Analysis & Primary Dimensions

Understanding the exact structural schema of the HDF5 file is critical for writing robust PyTorch training scripts. [cite: 1] The internal architecture reflects a land-seismic processing-software trace-header export rather than a raw SEG-Y dump. [cite: 1] The dataset is highly organized, and the traces are pre-sorted contiguously by SHOTID, which allows gather extraction via fast contiguous HDF5 slicing. [cite: 1]

2,424,923

TOTAL TRACES

907

UNIQUE SHOTS

1,501

SAMPLES / TRACE

- **Sample Rate & Timing:** The record length is 1500.0 ms (1.500 seconds) with a sample interval of 1000.0 μ s (1.000 ms). [cite: 1] This contradicts the 2ms conversion assumption found in standard pipelines, requiring a downstream fix for ms conversions. [cite: 1]

- **Gather Sizing:** Traces per shot gather vary significantly. [cite: 1] The minimum is 1 trace per shot, while the maximum is 2,685 traces. [cite: 1] The mean gather size is 2,673.6 traces with a standard deviation of 159.1. [cite: 1]
- **Primary Array:** The main seismic amplitudes are stored in the data_array dataset (shape: 2,424,923 × 1501, float32), consuming roughly 14.56 GB of space. [cite: 1]

3. Field Semantics & Metadata

Vendor seismic-processing exports often carry numerous header fields. [cite: 1] The following core fields are essential for grouping, geometry, and labeling within the data loaders: [cite: 1]

Category	Key Fields	Meaning / Usage
Gather Identifiers	SHOTID, CHANNEL, CDP	SHOTID groups traces into common-source gathers, which serves as the natural input unit for 2D U-Net models. [cite: 1]
Offset Geometry	OFFSET, OFFSET_FLT, AZIMUTH	Represents the source-receiver distance, which is the most critical variable for moveout and Quality Control (QC) plots. [cite: 1]
Quality Flags	TRTYPE, USE	Trace-type flags traditionally used for excluding unusable or dead traces. [cite: 1]
Static Corrections	SOURCE_STATIC, REC_STATIC	Indicates that a refraction-statics analysis has already been executed on this survey. [cite: 1]

4. Label Coverage & Break-Time Candidates

A crucial step in configuring supervised machine learning models is establishing the correct ground-truth labels. [cite: 1] The dataset contains three potential candidates for break-time labels. [cite: 1]

- **SPARE1:** This field acts as a generic spare field that has been repurposed to hold the actual picks. [cite: 1] It is populated with 1,211,857 labeled traces, giving it a label coverage of 49.98%. [cite: 1] Valid values range from -1.0 to 881.0, with a mean valid pick of 249.22. [cite: 1]
- **FIRST_BREAK_TIME:** Despite being an explicitly named field, it contains 0 labeled traces (0.00% coverage). [cite: 1]
- **MODELLED_BREAK_TIME:** Likely an independent theoretical refraction-statics estimate, this field is also entirely unpopulated (0.00% coverage). [cite: 1]

Label Reconciliation Conclusion: Because FIRST_BREAK_TIME and MODELLED_BREAK_TIME are completely empty, SPARE1 must be strictly utilized as the primary label field for generating the U-Net training masks. [cite: 1]

5. Data Quality, Statistics & Amplitude Distributions

Ensuring the dataset is mathematically clean prevents network training failures, such as NaN loss explosions during backward passes. [cite: 1] Streaming the trace-amplitude dataset yielded the following global statistics:

-0.0028

GLOBAL MEAN
AMPLITUDE

6.6097

STANDARD DEVIATION

0

NAN / INF VALUES

The total absence of NaN and Inf values within the data_array confirms that the mathematical integrity of the dataset is intact and ready for scaling experiments. [cite: 1] Global amplitudes range exactly from a minimum of -3284.134277 to a maximum of 3310.988525. [cite: 1]

5.1 Trace-Quality Flags vs. RMS Energy

While the dataset is mathematically valid, physical signal quality presents minor anomalies. [cite: 1] According to the TRTYPE header flag, 100.00% (all 2,424,923 traces) are explicitly marked as "Seismic (live)". [cite: 1] Similarly, the USE flag marks 100% of the traces as valid (value of 1). [cite: 1]

However, calculating the per-trace Root Mean Square (RMS) energy reveals conflicting realities regarding dead or low-energy traces: [cite: 1]

- **Dead Traces:** 381 traces (0.016%) possess near-zero RMS energy (less than a $1e-08$ threshold), meaning they are essentially all-zero traces. [cite: 1]
- **Low-Energy Traces:** 24,250 traces fall into the bottom 1st percentile of overall RMS energy. [cite: 1]

Actionable Insight: Because the TRTYPE flag incorrectly labels RMS-dead traces as 'live', the header flag alone is dangerously unreliable for filtering training data. [cite: 1] The custom PyTorch dataset loader must implement an independent RMS-based energy threshold check as an authoritative filter to safely exclude these 381 dead traces before they reach the neural network. [cite: 1]